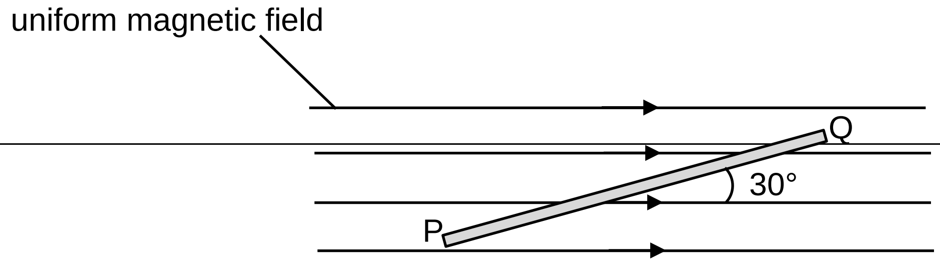
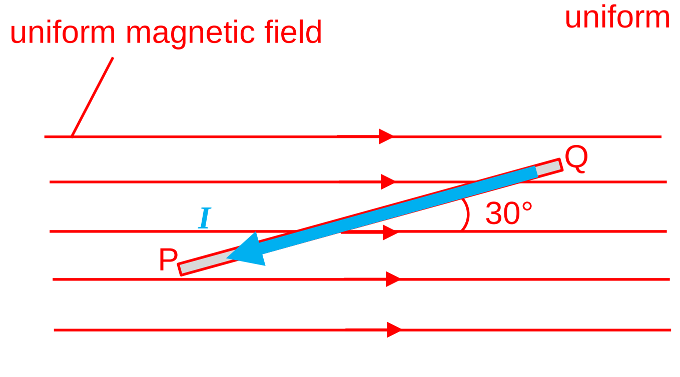
24	A horizontal wire PQ of length 0.50 m and weight 0.50 N is placed at an angle 30° to the uniform magnetic field as shown below. The wire is balanced by the magnetic force of the magnetic field of magnetic flux density of 1.0 T.



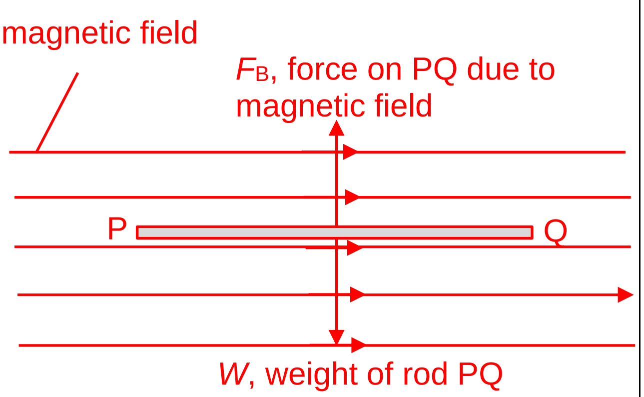
What is the magnitude and direction of the current in the wire?

	magnitude of current	direction of current
A	1.0 A	P to Q
B	2.0 A	P to Q
C	1.0 A	Q to P
D	2.0 A	Q to P

Answer: D



Top view



Side view

For the force on rod PQ to be in the upwards vertical direction, the direction of current must be from Q to P.

Considering the vertical forces,

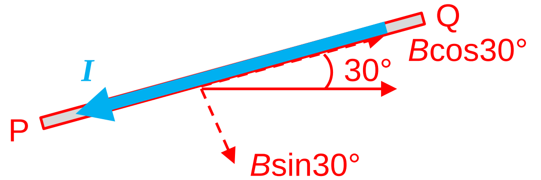
$F_B = W$

$BIl\sin\theta = mg$

$I = \dfrac{mg}{Bl\sin\theta}$

$I = \dfrac{0.50}{(1.0)(0.50)\sin 30^0}$

$I = 2.0$ A



Resolving the direction of the magnetic field

25 In Fig. 25.1 below, a beam of particles, each of charge q and mass m, is travelling at speed v